Pearson Edexcel Level 1/Level 2 – style practice paper

GCSE (9–1) Mathematics

1MA1/1H & 2H – Higher Tier

Topic 4: Geometry and measures

Centre Number		**Candidate Number**	
Surname		**Other names**	

Practice questions in the style of Pearson Edexcel GCSE Mathematics (1MA1). Every question is original; the layout, command words, mark allocations and mark scheme (M, A, B marks) follow the real exam format. Calculator rules are given in the instructions below.

Time allowed	About 145 minutes (roughly 1 minute per mark)
Instructions	• Use **black** ink or ball-point pen. • Answer **all** questions. • Answer the questions in the spaces provided – there may be more space than you need. • Calculators must **not** be used in Section A. Calculators may be used in Section B. • Diagrams are **NOT** accurately drawn, unless otherwise indicated. • You must **show all your working**. • **Unless stated, give non-exact answers to 3 significant figures.**
Information	• The total mark for this paper is **146**. • The marks for **each** question are shown in brackets – use this as a guide as to how much time to spend on each question. • Questions labelled with an asterisk (*) are ones where the quality of your written communication will be assessed.
Advice	• Read each question carefully before you start to answer it. • Keep an eye on the time. • Try to answer every question. • Check your answers if you have time at the end.

Name: _____ Date: _____ Mark: _____ / 146

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Section A: Non-calculator

You must not use a calculator for questions 1 to 20.

1 AB is parallel to CD. PQ is a straight line crossing both lines.

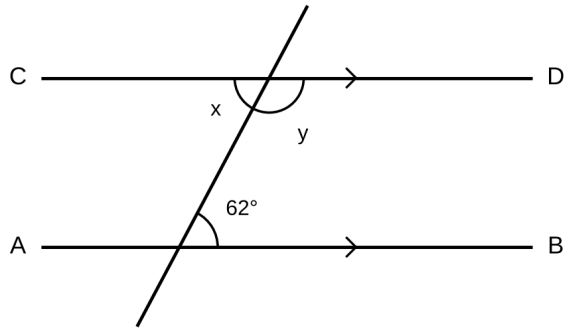


*Diagram **NOT** accurately drawn*

(a) Find the size of angle x. Give a reason for your answer. **(2)**

.....

.....

(b) Find the size of angle y. Give a reason for your answer. **(2)**

.....

.....

(Total for Question 1 is 4 marks)

2 (a) The interior angle of a regular polygon is 150°. Work out the number of sides of the polygon. **(3)**

.....

(b) The angles of a pentagon are $x°$, $(x + 20)°$, $2x°$, $110°$ and $130°$. Work out the value of x. **(3)**

.....

(c) Work out the size of one interior angle of a regular octagon. **(2)**

.....

(Total for Question 2 is 8 marks)

3 A ship sails from A to B on a bearing of 065°. It then sails from B to C on a bearing of 140°.

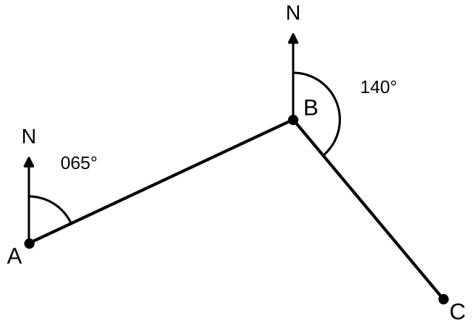


*Diagram **NOT** accurately drawn*

(a) Work out the bearing of A from B. **(2)**

.....

(b) Work out the size of angle ABC. **(2)**

.....

(Total for Question 3 is 4 marks)

4 A ladder of length 6.5 m leans against a vertical wall. The foot of the ladder is on horizontal ground, 2.5 m from the wall.

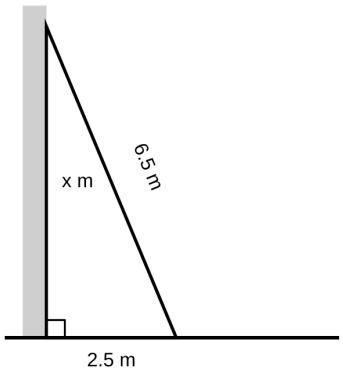


*Diagram **NOT** accurately drawn*

Work out how far up the wall the ladder reaches. **(3)**

.....

(Total for Question 4 is 3 marks)

5 (a) Write down the exact value of $\sin 30^\circ$. **(1)**

.....

(b) Write down the exact value of $\cos 60^\circ$. **(1)**

.....

(c) Write down the exact value of $\tan 60^\circ$. **(1)**

.....

(d) Write down the exact value of $\cos 30^\circ$. **(1)**

.....

(Total for Question 5 is 4 marks)

6 ABC is a right-angled triangle. $AB = 10$ cm and angle $A = 30°$.

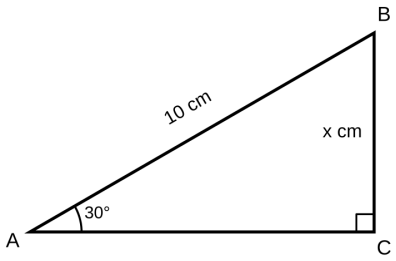


*Diagram **NOT** accurately drawn*

(a) Work out the exact value of x. **(2)**

.....

(b) Work out the exact length of AC. Give your answer as a surd. **(3)**

.....

(Total for Question 6 is 5 marks)

7 The diagram shows an L-shape. All the corners are right angles.

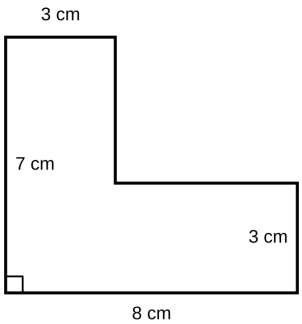


*Diagram **NOT** accurately drawn*

(a) Work out the area of the shape. **(3)**

.....

(b) Work out the perimeter of the shape. **(2)**

.....

(Total for Question 7 is 5 marks)

8 The diagram shows a sector of a circle of radius 5 cm. The angle of the sector is 72°.

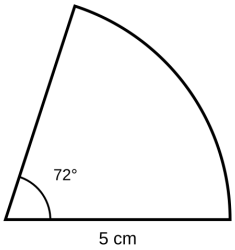


*Diagram **NOT** accurately drawn*

(a) Work out the area of the sector. Give your answer in terms of π. **(3)**

.....

(b) Work out the perimeter of the sector. Give your answer in terms of π. **(3)**

.....

(Total for Question 8 is 6 marks)

9 The diagram shows a prism. The cross-section is a right-angled triangle with sides 6 cm, 8 cm and 10 cm. The length of the prism is 10 cm.

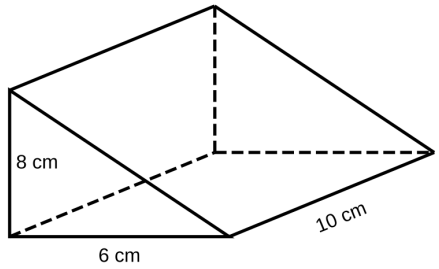


*Diagram **NOT** accurately drawn*

(a) Work out the volume of the prism. **(2)**

.....

(b) Work out the total surface area of the prism. **(3)**

.....

(Total for Question 9 is 5 marks)

10 An ice cream is a cone with a hemisphere on top. The cone has radius 3 cm and height 9 cm. The radius of the hemisphere is 3 cm.

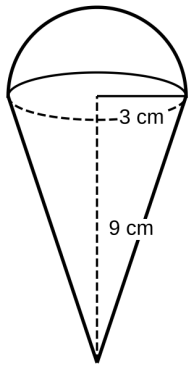


*Diagram **NOT** accurately drawn*

Work out the total volume of the ice cream. Give your answer in terms of π. **(4)**

.....

(Total for Question 10 is 4 marks)

11 The two triangles are mathematically similar.

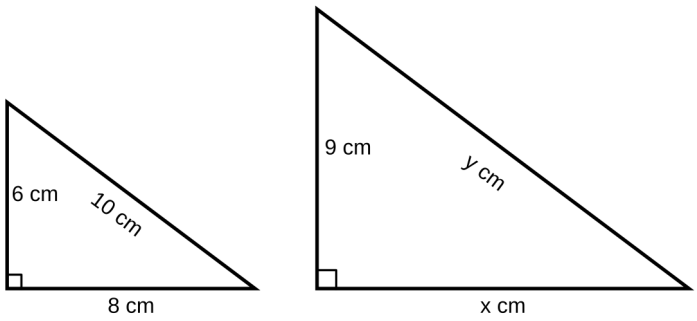


*Diagram **NOT** accurately drawn*

(a) Work out the value of x. **(2)**

.....

(b) Work out the value of y. **(2)**

.....

(Total for Question 11 is 4 marks)

12 (a) Two solid shapes are mathematically similar. The heights of the shapes are 6 cm and 9 cm. The volume of the smaller shape is 80 cm³. Work out the volume of the larger shape. **(3)**

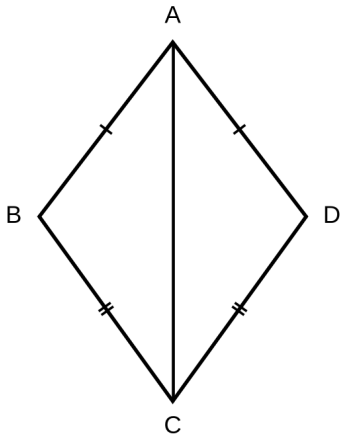
.....

(b) Two similar shapes have areas of 36 cm² and 81 cm². The perimeter of the smaller shape is 28 cm. Work out the perimeter of the larger shape. **(3)**

.....

(Total for Question 12 is 6 marks)

13 $ABCD$ is a kite with $AB = AD$ and $CB = CD$.



(a) Prove that triangle ABC is congruent to triangle ADC. **(3)**

.....

.....

.....

.....

(b) Hence explain why angle ABC equals angle ADC. **(1)**

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.....

(Total for Question 13 is 4 marks)

14 A, B and C are points on the circle with centre O. Angle $AOB = 124°$.

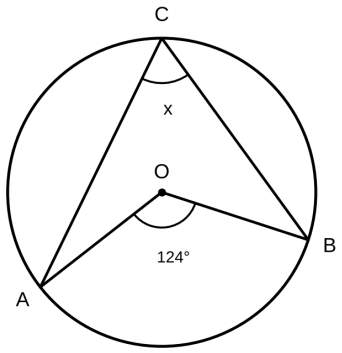


*Diagram **NOT** accurately drawn*

Work out the size of angle x. Give a reason for your answer. **(2)**

.....

.....

(Total for Question 14 is 2 marks)

15 $ABCD$ is a cyclic quadrilateral.

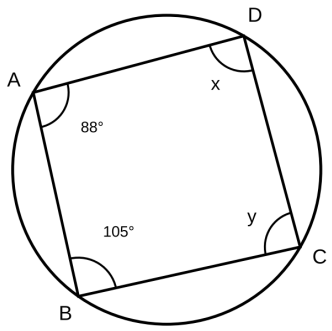


*Diagram **NOT** accurately drawn*

(a) Work out the size of angle x. Give a reason for your answer. **(2)**

.....

.....

(b) Work out the size of angle y. **(2)**

.....

(Total for Question 15 is 4 marks)

16 TA and TB are tangents to the circle with centre O. Angle $ATB = 44°$.

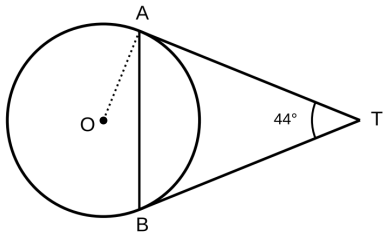


*Diagram **NOT** accurately drawn*

(a) Work out the size of angle TAB. Give a reason for each step of your working. **(3)**

.....

.....

.....

.....

(b) Work out the size of angle OAB. **(2)**

.....

.....

(Total for Question 16 is 5 marks)

17 OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. M is the midpoint of AB. C is the point on OB such that $OC : CB = 1 : 2$.

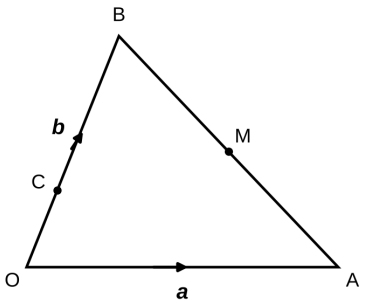


*Diagram **NOT** accurately drawn*

(a) Write $\overrightarrow{AB}$ in terms of **a** and **b**. **(1)**

.....

(b) Find $\overrightarrow{OM}$ in terms of **a** and **b**. Give your answer in its simplest form. **(2)**

.....

(c) Find $\overrightarrow{AC}$ in terms of **a** and **b**. **(2)**

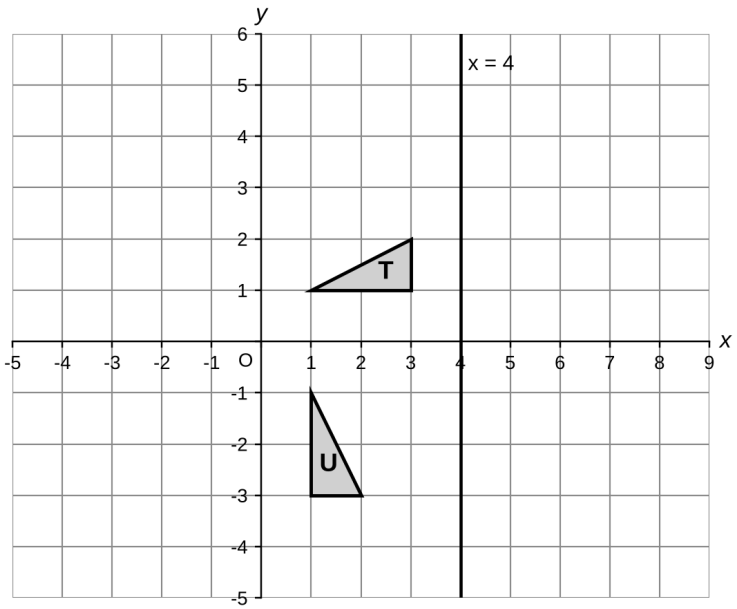
.....

(d) Find $\overrightarrow{CM}$ in terms of **a** and **b**. **(3)**

.....

(Total for Question 17 is 8 marks)

18 The grid shows triangle *T* and triangle *U*.



(a) Describe fully the single transformation that maps triangle *T* onto triangle *U*. **(3)**

.....

.....

.....

(b) Reflect triangle T in the line $x = 4$. Label the image V. **(2)**

(c) Enlarge triangle T by scale factor 2 with centre $(0, 0)$. Label the image W. **(2)**

(d) Triangle X has vertices at $(-1, -1)$, $(-3, -1)$ and $(-3, -2)$. Describe fully the single transformation that maps triangle T onto triangle X. **(3)**

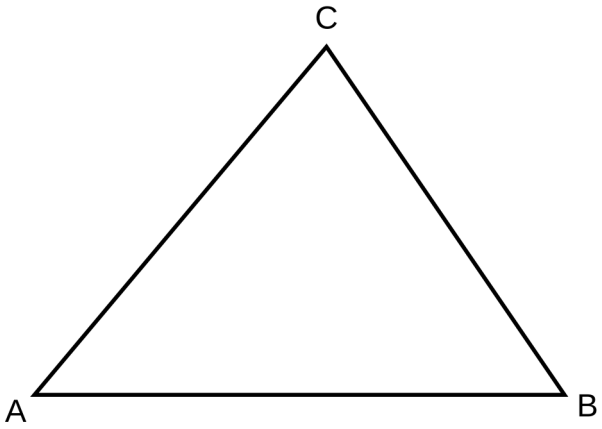
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(Total for Question 18 is 10 marks)

19 Triangle ABC is drawn accurately. The scale is 1 cm to 1 cm.



(a) Using ruler and compasses, construct the bisector of angle BAC. You must show all your construction lines. **(2)**

(b) Shade the region inside the triangle that is **both** nearer to AB than to AC **and** less than 5 cm from A. **(3)**

(Total for Question 19 is 5 marks)

20 (a) The points A and B have coordinates $(-2, 3)$ and $(4, -5)$. Work out the length of AB. **(3)**

.....

(b) Find the coordinates of the midpoint of AB. **(2)**

.....

(Total for Question 20 is 5 marks)

Section B: Calculator

A calculator may be used for questions 21 to 30.

21 A ship sails 5 km from A to B on a bearing of 065°. It then sails 4 km from B to C on a bearing of 140°. (Angle $ABC = 105°$.)

Work out the distance AC. Give your answer correct to 3 significant figures. **(3)**

.....

(Total for Question 21 is 3 marks)

22 A tower stands on horizontal ground. From point A, 45 m from the base of the tower, the angle of elevation of the top of the tower is 28°.

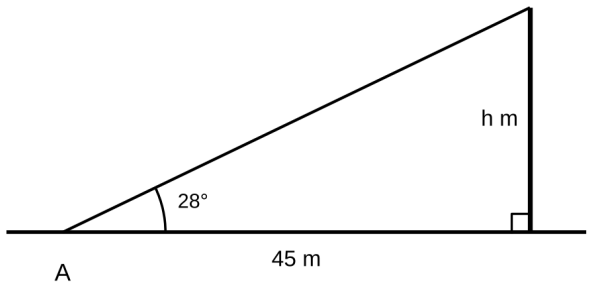


*Diagram **NOT** accurately drawn*

Work out the height, h m, of the tower. Give your answer correct to 3 significant figures. **(3)**

.....

(Total for Question 22 is 3 marks)

23 In triangle ABC, $AB = 7$ cm, $AC = 9$ cm and angle $BAC = 65°$.

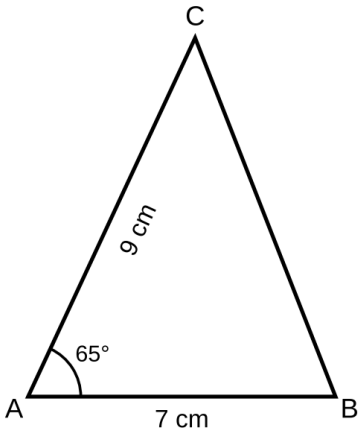


*Diagram **NOT** accurately drawn*

(a) Work out the length of BC. Give your answer correct to 3 significant figures. **(3)**

.....

(b) Work out the area of triangle ABC. Give your answer correct to 3 significant figures. **(2)**

.....

(Total for Question 23 is 5 marks)

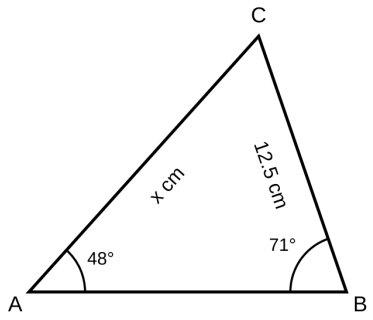


*Diagram **NOT** accurately drawn*

24 In triangle ABC, angle $A = 48°$, angle $B = 71°$ and $BC = 12.5$ cm. Work out the length of AC. Give your answer correct to 3 significant figures. **(3)**

.....

(Total for Question 24 is 3 marks)

25 The diagram shows a cuboid $ABCDEFGH$. $AB = 12$ cm, $BC = 5$ cm and $AE = 4$ cm.

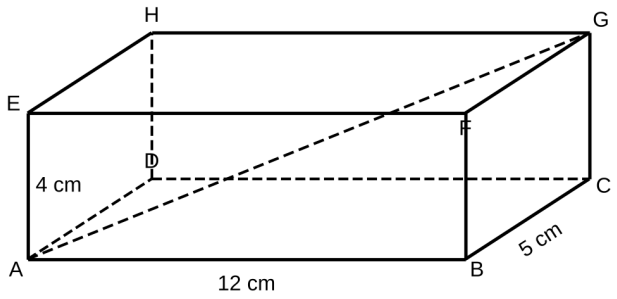


*Diagram **NOT** accurately drawn*

(a) Work out the length of AG. Give your answer correct to 3 significant figures. **(3)**

.....

(b) Work out the size of the angle between AG and the base $ABCD$. Give your answer correct to 1 decimal place. **(3)**

.....

(Total for Question 25 is 6 marks)

26 A sector of a circle has radius 8.4 cm and angle 135°.

(a) Work out the area of the sector. Give your answer correct to 3 significant figures. **(3)**

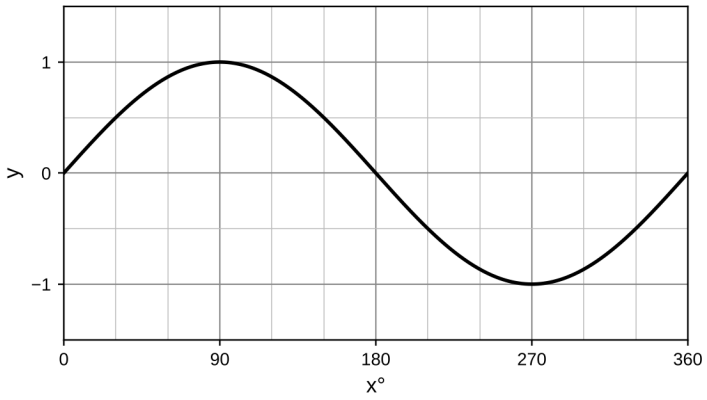
.....

(b) Work out the perimeter of the sector. Give your answer correct to 3 significant figures. **(3)**

.....

(Total for Question 26 is 6 marks)

27 Here is a sketch of the graph of $y = \sin x^\circ$ for $0 \leq x \leq 360$.



(a) Use the graph to find all the solutions of $\sin x^\circ = 0.5$ in the range $0 \leq x \leq 360$. **(2)**

.....

(b) Find all the solutions of $\sin x^\circ = -0.5$ in the range $0 \leq x \leq 360$. **(2)**

.....

(Total for Question 27 is 4 marks)

28 PQR is a right-angled triangle with the right angle at Q. $PQ = 7.5$ cm and $QR = 11.2$ cm.

(a) Work out the size of angle QPR. Give your answer correct to 1 decimal place. **(3)**

.....

(b) Work out the length of PR. Give your answer correct to 3 significant figures. **(2)**

.....

(Total for Question 28 is 5 marks)

29 (a) A solid pyramid has a square base of side 8 cm and a perpendicular height of 9 cm. Work out the volume of the pyramid. **(2)**

.....

(b) A solid hemisphere has radius 6 cm. Work out its total surface area. Give your answer in terms of π. **(3)**

.....

(Total for Question 29 is 5 marks)

30 A cylinder has radius 4 cm and height 10 cm.

(a) Work out the total surface area of the cylinder. Give your answer correct to 3 significant figures. **(3)**

.....

(b) Work out the volume of the cylinder. Give your answer in terms of π. **(2)**

.....

(Total for Question 30 is 5 marks)

TOTAL FOR PAPER IS 146 MARKS

END

MARK SCHEME – Topic 4: Geometry and measures

Guidance on marking: M marks are method marks, awarded for a correct method (even if the final answer is wrong). **A** marks are accuracy marks, awarded only if the relevant M mark has been gained. **B** marks are independent marks (no method needed). **C** marks are communication marks. **P** marks are process marks (a correct approach to a problem). **ft** = follow through from a previous incorrect answer. **oe** = or equivalent. **cao** = correct answer only. **SC** = special case. If the answer on the answer line differs from the working, the answer line is marked unless the working is clearly the intended answer. Correct answers with no working score full marks unless the question says 'You must show your working'.

Question	Working	Answer	Mark	Notes
1(a)	B1 $x = 62$ C1 alternate angles are equal	62° because ...	2 AO2	
1(b)	B1 $y = 118$ C1 co-interior (allied) angles add to 180° / angles on a straight line	118° because ...	2 AO2	
2(a)	M1 exterior angle $= 180 - 150 = 30$ M1 $360 \div 30$ A1 12	12	3 AO1	
2(b)	B1 sum of interior angles $= 540°$ M1 $4x + 20 + 240 = 540$ A1 $x = 70$	x = 70	3 AO3	
2(c)	M1 $360 \div 8 = 45$ or $(8 - 2) \times 180 \div 8$ A1 135	135°	2 AO1	
3(a)	M1 $65 + 180$ or alternate angle $65°$ at B A1 245	245°	2 AO2	
3(b)	M1 $245 - 140$ A1 105	105°	2 AO3	
4	M1 $x^2 + 2.5^2 = 6.5^2$ M1 $x^2 = 42.25 - 6.25 = 36$ A1 6	6 m	3 AO3	
5(a)	B1 $\frac{1}{2}$	$\frac{1}{2}$	1 AO1	
5(b)	B1 $\frac{1}{2}$	$\frac{1}{2}$	1 AO1	
5(c)	B1 $\sqrt{3}$	$\sqrt{3}$	1 AO1	
5(d)	B1 $\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	1 AO1	
6(a)	M1 $x = 10\sin 30$ A1 5	5	2 AO1	
6(b)	M1 $AC = 10\cos 30$ M1 $10 \times \frac{\sqrt{3}}{2}$ A1 $5\sqrt{3}$	$5\sqrt{3}$ cm	3 AO1	
7(a)	M1 splits shape correctly e.g. $7 \times 3 + 5 \times 3$ M1 $21 + 15$ A1 36	36 cm²	3 AO1	

Question	Working	Answer	Mark	Notes
7(b)	M1 finds missing lengths 5 and 4 A1 30	30 cm	2 AO1	
8(a)	M1 $\frac{72}{360}$ M1 $\frac{72}{360} \times \pi \times 5^2$ A1 5π	5π cm²	3 AO1	
8(b)	M1 arc $= \frac{72}{360} \times 2\pi \times 5$ M1 $2\pi + 5 + 5$ A1 $10 + 2\pi$	$10 + 2\pi$ cm	3 AO1	
9(a)	M1 $\frac{1}{2} \times 6 \times 8 \times 10$ A1 240	240 cm³	2 AO1	
9(b)	M1 triangle area $= 24$ (two of them) M1 $(6 + 8 + 10) \times 10 = 240$ A1 288	288 cm²	3 AO1	
10	M1 cone $\frac{1}{3}\pi \times 3^2 \times 9$ $(= 27\pi)$ M1 hemisphere $\frac{2}{3}\pi \times 3^3$ $(= 18\pi)$ M1 $27\pi + 18\pi$ A1 45π	45π cm³	4 AO3	
11(a)	M1 scale factor $\frac{9}{6} = 1.5$ A1 12	12	2 AO2	
11(b)	M1 10×1.5 A1 15	15	2 AO1	
12(a)	M1 linear scale factor $\frac{9}{6} = \frac{3}{2}$ M1 $80 \times \left(\frac{3}{2}\right)^3$ A1 270	270 cm³	3 AO3	
12(b)	M1 $\sqrt{\frac{81}{36}} = \frac{9}{6}$ M1 $28 \times \frac{3}{2}$ A1 42	42 cm	3 AO3	
13(a)	B1 $AB = AD$ and $CB = CD$ (given) B1 AC is common C1 so the triangles are congruent (SSS)		3 AO2	
13(b)	C1 corresponding angles in congruent triangles are equal		1 AO2	
14	B1 62 C1 the angle at the centre is twice the angle at the circumference	62° because ...	2 AO2	
15(a)	B1 $x = 75$ C1 opposite angles of a cyclic quadrilateral add up to 180°	75° because ...	2 AO2	
15(b)	M1 $180 - 88$ A1 92	92°	2 AO1	

Question	Working	Answer	Mark	Notes
16(a)	M1 $TA = TB$ (tangents from a point are equal) so triangle TAB is isosceles M1 $(180 - 44) \div 2$ A1 68	68°	3 AO2	
16(b)	M1 angle $OAT = 90°$ (radius meets tangent at right angle) A1 $90 - 68 = 22$	22°	2 AO2	
17(a)	B1 $\mathbf{b} - \mathbf{a}$	$\mathbf{b} - \mathbf{a}$	1 AO1	
17(b)	M1 $\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a})$ A1 $\frac{1}{2}(\mathbf{a} + \mathbf{b})$	$\frac{1}{2}(\mathbf{a} + \mathbf{b})$	2 AO1	
17(c)	M1 $\overrightarrow{AO} + \overrightarrow{OC}$ or $-\mathbf{a} + \frac{1}{3}\mathbf{b}$ A1 $\frac{1}{3}\mathbf{b} - \mathbf{a}$	$\frac{1}{3}\mathbf{b} - \mathbf{a}$	2 AO1	
17(d)	M1 $\overrightarrow{CM} = \overrightarrow{CO} + \overrightarrow{OM}$ M1 $-\frac{1}{3}\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$ A1 $\frac{1}{2}\mathbf{a} + \frac{1}{6}\mathbf{b}$	$\frac{1}{2}\mathbf{a} + \frac{1}{6}\mathbf{b}$	3 AO1	
18(a)	B1 rotation B1 90° clockwise B1 about the origin $(0, 0)$		3 AO2	
18(b)	M1 any vertex correctly reflected or correct orientation A1 triangle with vertices $(7, 1)$, $(5, 1)$, $(5, 2)$		2 AO1	
18(c)	M1 enlargement by 2 with 2 correct vertices (or correct shape wrong centre) A1 vertices $(2, 2)$, $(6, 2)$, $(6, 4)$		2 AO1	
18(d)	B1 rotation B1 180° B1 about the origin $(0, 0)$		3 AO2	
19(a)	M1 arcs centred on A cutting AB and AC, then equal arcs from those points A1 correct bisector (±2°) with construction arcs shown		2 AO2	
19(b)	B1 uses the angle bisector from part (a) M1 arc of radius 5 cm centred on A A1 correct region shaded: between AB and the bisector, inside the arc		3 AO3	
20(a)	M1 $(4 - (-2))^2 + (-5 - 3)^2$ M1 $\sqrt{36 + 64}$ A1 10	10	3 AO1	
20(b)	M1 $\left(\frac{-2 + 4}{2}, \frac{3 + (-5)}{2}\right)$ A1 $(1, -1)$	(....., ...)	2 AO1	
21	M1 $AC^2 = 5^2 + 4^2 - 2 \times 5 \times 4 \times \cos 105°$ M1 $AC^2 = 51.35\ldots$ A1 7.17	7.17 km	3 AO3	

Question	Working	Answer	Mark	Notes
22	M1 $\tan 28 = \frac{h}{45}$ M1 $h = 45 \times \tan 28$ A1 23.9	23.9 m	3 AO3	
23(a)	M1 $BC^2 = 7^2 + 9^2 - 2 \times 7 \times 9\cos 65°$ M1 $\sqrt{76.75...}$ A1 8.76	8.76 cm	3 AO1	
23(b)	M1 $\frac{1}{2} \times 7 \times 9 \times \sin 65°$ A1 28.5	28.5 cm²	2 AO1	
24	M1 $\frac{x}{\sin 71°} = \frac{12.5}{\sin 48°}$ M1 $x = \frac{12.5\sin 71°}{\sin 48°}$ A1 15.9	15.9 cm	3 AO1	
25(a)	M1 $AC^2 = 12^2 + 5^2$ $(AC = 13)$ M1 $AG^2 = 13^2 + 4^2$ $(= 185)$ A1 13.6	13.6 cm	3 AO3	
25(b)	M1 $\tan\theta = \frac{4}{13}$ (or $\frac{4}{AC}$) M1 $\tan^{-1}\left(\frac{4}{13}\right)$ A1 17.1	17.1°	3 AO3	
26(a)	M1 $\frac{135}{360} \times \pi \times 8.4^2$ M1 $= 83.12...$ A1 83.1	83.1 cm²	3 AO1	
26(b)	M1 arc $= \frac{135}{360} \times 2\pi \times 8.4$ $(= 19.79...)$ M1 $19.79 + 8.4 + 8.4$ A1 36.6	36.6 cm	3 AO1	
27(a)	B1 30 B1 150	x = ... or x = ...	2 AO2	
27(b)	B1 210 B1 330	x = ... or x = ...	2 AO2	
28(a)	M1 $\tan P = \frac{11.2}{7.5}$ M1 $\tan^{-1}\left(\frac{11.2}{7.5}\right)$ A1 56.2	56.2°	3 AO1	
28(b)	M1 $\sqrt{7.5^2 + 11.2^2}$ A1 13.5	13.5 cm	2 AO1	
29(a)	M1 $\frac{1}{3} \times 8^2 \times 9$ A1 192	192 cm³	2 AO1	
29(b)	M1 curved surface $\frac{1}{2} \times 4\pi \times 6^2 = 72\pi$ M1 flat face $\pi \times 6^2 = 36\pi$ A1 108π	108π cm²	3 AO1	

Question	Working	Answer	Mark	Notes
30(a)	M1 $2\pi \times 4 \times 10$ (= 251.3…) M1 $+2 \times \pi \times 4^2$ (= 100.5…) A1 352	352 cm²	3 AO1	
30(b)	M1 $\pi \times 4^2 \times 10$ A1 160π	160π cm³	2 AO1	

Where marks are lost in Topic 4

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Angle reasons:** give the full name (alternate, co-interior, angles in a triangle, angle at the centre…). Say *why*, not just the number.
- ☐ **Polygons:** exterior angles add to 360°; interior + exterior = 180°; interior angle sum $= (n-2) \times 180^\circ$.
- ☐ **Bearings:** measure clockwise from north, always three figures. Draw a north line at *each* point. Back-bearing = ±180°.
- ☐ **Exact trig:** learn $\sin 30 = \frac{1}{2}$, $\cos 30 = \frac{\sqrt{3}}{2}$, $\tan 30 = \frac{1}{\sqrt{3}}$, $\tan 45 = 1$, $\sin 60 = \frac{\sqrt{3}}{2}$, $\cos 60 = \frac{1}{2}$, $\tan 60 = \sqrt{3}$.
- ☐ **Similar shapes:** area scale factor = (length scale factor)², volume = (length scale factor)³. Convert the ratio you are given first.
- ☐ **Circle theorems:** angle at centre = 2 × angle at circumference; angles in the same segment equal; opposite angles in a cyclic quadrilateral add to 180°; radius □ tangent; tangents from a point are equal.
- ☐ **Vectors:** go from the start to the end by any route; reverse a vector by changing its sign. Simplify to a single bracket.
- ☐ **Transformations:** always state the *type* plus all details: rotation (angle, direction, centre), reflection (equation of line), enlargement (scale factor, centre), translation (column vector).
- ☐ **Sine/cosine rule:** sine rule for two angles + a side; cosine rule for two sides and the angle between (SAS) or three sides (SSS). Check your calculator is in degrees.
- ☐ **3D problems:** find the base diagonal first, then use it in a second right-angled triangle.